

Power BI Assignment 1 – Data Transformation & Data Modeling

Screenshots of the solution and the data model view

Brief explanation of transformation and modeling logic

1. Dataset import screenshots

1.1. Import List of Orders.csv into Power BI.

The screenshot shows the Power BI Desktop interface with the 'Get Data (Power Query)' dialog box open. The dialog box is titled 'Preview file data' and shows the file path: C:\Users\SHABARLA\Documents\Data Analytics\Assignment\Module 2-PowerBI>List of Orders.csv. The file origin is set to '65001: Unicode (UTF-8)', the delimiter is 'Comma', and the data type detection is 'Based on entire data set'. A preview of the data is shown in a table with 5 columns: Column1, Column2, Column3, Column4, and Column5. The data is truncated at the bottom.

Column1	Column2	Column3	Column4	Column5
B-25785	15-09-2018	Ayush	West Bengal	Kolkata
B-25786	19-09-2018	Abhishek	Karnataka	Bangalore
B-25787	20-09-2018	Asish	Jammu and Kash...	Kashmir
B-25788	21-09-2018	Dinesh	Tamil Nadu	Chennai
B-25789	22-09-2018	Akshay	Uttar Pradesh	Lucknow
B-25790	23-09-2018	Sajal	Bihar	Patna
B-25791	24-09-2018	Avish	Kerala	Thiruvananthapura...
B-25792	24-09-2018	Abhishek	Maharashtra	Mumbai
B-25793	24-09-2018	Siddharth	Madhya Pradesh	Indore
B-25794	24-09-2018	Aditya	Himachal Pradesh	Simla
B-25795	24-09-2018	Sukant	Sikkim	Gangtok
B-25796	24-09-2018	Sukrith	Maharashtra	Mumbai
B-25797	30-09-2018	Saaptik	Madhya Pradesh	Indore
B-25798	01-10-2018	Shishu	Andhra Pradesh	Hyderabad
B-25799	01-10-2018	Divyansh	Gujarat	Ahmedabad

The data in the preview may have been truncated due to size limits.

Buttons: Back, Cancel, Add table using examples, Load

Untitled - Power BI Desktop

Sign in

File Home Help Table tools Column tools

Paste Cut Copy

Get data Excel OneLake SQL Server Enter data Data Lake Recent sources

Transform data Refresh

Manage relationships

New measure Quick measure New column New table

Manage roles View as

Sensitivity

Publish

Clipboard

Queries

Relationships

Calculations

Security

Sensitivity

Share

Column1 Column2 Column3 Column4 Column5

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur

Table: List of Orders (561 rows) Column: Column1 (502 distinct values)

Search

List of Orders

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1.2. Open the list of orders in the Power Query Editor using the Transform Data option.

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Data Type: Text Detect Data Type Rename

Any Column

Split Column Format Merge Columns Extract Parse

Text Column

Statistics Standard Scientific Trigonometry Rounding Information

Number Column

Date Time Duration

Run R script Run Python script

Scripts

Queries [1]

List of Orders

Order Date CustomerName State City

31-01-2019 Nirja Delhi Delhi

388 31-01-2019 Nirja Delhi Delhi

389 01-02-2019 Anjali Delhi Delhi

390 02-02-2019 Mugdha Delhi Delhi

391 03-02-2019 Mansi Madhya Pradesh Indore

392 03-02-2019 Gaurav Rajasthan Udaipur

393 03-02-2019 Harshal Delhi Delhi

394 03-02-2019 Omkar Delhi Delhi

395 04-02-2019 Yohann Delhi Delhi

396 04-02-2019 Prashant Delhi Delhi

397 04-02-2019 Yogesh Madhya Pradesh Indore

398 04-02-2019 Anmol Rajasthan Udaipur

399 05-02-2019 Diwakar Delhi Delhi

400 06-02-2019 Shubham Delhi Delhi

401

5 COLUMNS, 560 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 02:32 PM

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

first row as headers

Changed Type

1.3. Import Order Details.csv and Sales Target.csv into Power Query Editor.

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Reduce Rows Sort Split Column Group By Replace Values Combine

Queries [3]

- List of Orders
- Order Details
- Sales target

fx = Table.TransformColumnTypes("#Promoted headers", {"Order ID", type text},

	Order ID	Amount	Profit	Quantity	Category
1	B-25601	1275	-1148	7	Furn
2	B-25601	66	-12	5	Cloti
3	B-25601	8	-2	3	Cloti
4	B-25601	80	-56	4	Elect
5	B-25602	168	-111	2	Elect
6	B-25602	424	-272	5	Elect
7	B-25602	2617	1151	4	Elect
8	B-25602	561	212	3	Cloti
9	B-25602	119	-5	8	Cloti
10	B-25603	1355	-60	5	Cloti
11	B-25603	24	-30	1	Furn
12	B-25603	193	-166	3	Cloti
13	B-25603	180	5	3	Cloti
14	B-25603	116	16	4	Cloti

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

- Source
- Promoted headers
- Changed column type

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Reduce Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Combine

Queries [3]

- List of Orders
- Order Details
- Sales target

Order ID Amount Profit Quantity Category

Order ID	Amount	Profit	Quantity	Category
1 B-25601	1275	-1148	7	Furn
2 B-25601	66	-12	5	Clotl
3 B-25601	8	-2	3	Clotl
4 B-25601	80	-56	4	Elect
5 B-25602	168	-111	2	Elect
6 B-25602	424	-272	5	Elect
7 B-25602	2617	1151	4	Elect
8 B-25602	561	212	3	Clotl
9 B-25602	119	-5	8	Clotl
10 B-25603	1355	-60	5	Clotl
11 B-25603	24	-30	1	Furn
12 B-25603	193	-166	3	Clotl
13 B-25603	180	5	3	Clotl
14 B-25603	116	16	4	Clotl

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

- Source
- Promoted headers
- Changed column type

The three CSV files (**List of Orders**, **Order Details**, and **Sales Target**) were imported into Power BI using the **Get Data** feature. All datasets were opened in **Power Query Editor** to perform data cleansing and transformation before loading them into the data model.

2. Row Limiting & Data Types

2.1. Restrict the List of Orders table to the **first 500 rows**

The screenshot shows the Power BI Desktop interface with a table named 'List of Orders' containing 560 rows and 5 columns: Order ID, Order Date, CustomerName, State, and City. A dialog box titled 'Keep Top Rows' is open, prompting the user to specify how many rows to keep. The 'Number of rows' field is set to 500. The background table shows the first 24 rows of data.

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603				
B-25604				
B-25605				
B-25606				
B-25607				
B-25608				
B-25609				
B-25610				
B-25611				
B-25612				
B-25613				
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Bengal	Kolkata
B-25624	22-04-2018	Vini	Karnataka	Bangalore

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File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Merge Queries Append Queries Combine Files Combine

Queries [3] List of Orders Order Details Sales target

fx = Table.FirstN("#Changed Type",500)

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

first row as headers

Changed Type

Kept First 500 Rows

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2.2. Set the **Order Date** column to the **Date** data type.

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File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Date Use First Row as Headers Merge Queries Append Queries Combine Files Combine

Queries [3] List of Orders Order Details Sales target

fx = Table.TransformColumnTypes(#"Kept First 500 Rows",{"Order Date", type date})

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

first row as headers

Changed Type

Kept First 500 Rows

OrderDate changed to DATE ...

2.3. Change the Amount and Target columns to fixed decimal numbers.

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Data Sources Manage Parameters Parameters Export query results Output Data Refresh Preview Query Properties Advanced Editor Manage Choose Columns Manage Columns Remove Columns Remove Rows Keep Rows Remove Rows Sort Split Column Group By Data Type: Fixed decimal number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [3] List of Orders Order Details Sales target

fx = Table.TransformColumnTypes(#"Changed column type",{{"Amount", Currency.Type}, {"Profit", Currency.Type}})

	Order ID	\$ Amount	\$ Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1,148.00	7	Furniture	Bookcases
2	B-25601	66.00	-12.00	5	Clothing	Stole
3	B-25601	8.00	-2.00	3	Clothing	Hankerchief
4	B-25601	80.00	-56.00	4	Electronics	Electronic G
5	B-25602	168.00	-111.00	2	Electronics	Phones
6	B-25602	424.00	-272.00	5	Electronics	Phones
7	B-25602	2,617.00	1,151.00	4	Electronics	Phones
8	B-25602	561.00	212.00	3	Clothing	Saree
9	B-25602	119.00	-5.00	8	Clothing	Saree
10	B-25603	1,355.00	-60.00	5	Clothing	Trousers
11	B-25603	24.00	-30.00	1	Furniture	Chairs
12	B-25603	193.00	-166.00	3	Clothing	Saree
13	B-25603	180.00	5.00	3	Clothing	Trousers
14	B-25603	116.00	16.00	4	Clothing	Stole
15	B-25603	107.00	36.00	6	Clothing	Stole
16	B-25603	12.00	1.00	2	Clothing	Hankerchief
17	B-25603	38.00	18.00	1	Clothing	Kurti
18	B-25604	65.00	17.00	2	Clothing	T-shirt
19	B-25604	157.00	5.00	9	Clothing	Saree
20	B-25605	75.00	0.00	7	Clothing	Saree
21	B-25606	87.00	4.00	2	Clothing	Shirt
22	B-25607	50.00	15.00	4	Clothing	Leggings
23	B-25608	1,364.00	-1,864.00	5	Furniture	Tables
24						

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted headers

Changed column type

Amount Changed Type to Fix...

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 03:08 PM

Type here to search 38°C Sunny 03:08 PM 13-07-2026

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Data Sources Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Fixed decimal number Use First Row as Headers Merge Queries Append Queries Combine Files Combine

Queries [3] List of Orders Order Details Sales target

Month of Order Date Category Target

1	18-04-2026	Furniture	10,400.00
2	18-05-2026	Furniture	10,500.00
3	18-06-2026	Furniture	10,600.00
4	18-07-2026	Furniture	10,800.00
5	18-08-2026	Furniture	10,900.00
6	18-09-2026	Furniture	11,000.00
7	18-10-2026	Furniture	11,100.00
8	18-11-2026	Furniture	11,300.00
9	18-12-2026	Furniture	11,400.00
10	19-01-2026	Furniture	11,500.00
11	19-02-2026	Furniture	11,600.00
12	19-03-2026	Furniture	11,800.00
13	18-04-2026	Clothing	12,000.00
14	18-05-2026	Clothing	12,000.00
15	18-06-2026	Clothing	12,000.00
16	18-07-2026	Clothing	14,000.00
17	18-08-2026	Clothing	14,000.00
18	18-09-2026	Clothing	14,000.00
19	18-10-2026	Clothing	16,000.00
20	18-11-2026	Clothing	16,000.00
21	18-12-2026	Clothing	16,000.00
22	19-01-2026	Clothing	16,000.00
23	19-02-2026	Clothing	16,000.00
24	19-03-2026	Clothing	16,000.00

3 COLUMNS, 36 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

Sales target

APPLIED STEPS

Source

Promoted headers

Changed column type

Target Changed fixed decimal...

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The **List of Orders** table was restricted to the first **500 rows** to meet the assignment requirement. The **Order Date** column was converted to the **Date** data type to support accurate time-based analysis. The **Amount** and **Target** columns were converted to **Fixed Decimal Number** to ensure consistent numerical calculations.

3. Text Formatting

3.1. Format the CustomerName column to proper case to ensure consistent capitalization

The screenshot displays the Power BI Desktop interface for a query named "List of Orders". The data table has the following columns: Order ID, Order Date, CustomerName, State, and City. The CustomerName column is currently being edited with the formula: `= Table.TransformColumns(#"OrderDate changed to DATE data type",{{"CustomerName", Text.Proper, type text}})`. The right-hand pane shows the "Query Settings" for "List of Orders", including the "APPLIED STEPS" list: Source, first row as headers, Changed Type, Kept First 500 Rows, OrderDate changed to DATE ..., and Customer Name Capitalized E... (highlighted). The status bar at the bottom indicates "5 COLUMNS, 500 ROWS" and "Column profiling based on top 1000 rows".

Order ID	Order Date	CustomerName	State	City
1	01-04-2018	Bharat	Gujarat	Ahmedabad
2	01-04-2018	Pearl	Maharashtra	Pune
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	03-04-2018	Divsha	Rajasthan	Jaipur
5	05-04-2018	Kasheen	West Bengal	Kolkata
6	06-04-2018	Hazel	Karnataka	Bangalore
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	09-04-2018	Yogesh	Bihar	Patna
11	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	12-04-2018	Shrichand	Punjab	Chandigarh
13	12-04-2018	Mukesh	Haryana	Chandigarh
14	13-04-2018	Vandana	Himachal Pradesh	Simla
15	15-04-2018	Bhavna	Sikkim	Gangtok
16	15-04-2018	Kanak	Goa	Goa
17	17-04-2018	Sagar	Nagaland	Kohima
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	20-04-2018	Sarita	Maharashtra	Pune
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	22-04-2018	Monisha	Rajasthan	Jaipur
23	22-04-2018	Atharv	West Bengal	Kolkata
24	22-04-2018	Vini	Karnataka	Bangalore

4. Column Creation:

- 4.1. Merge the City and State columns to create a new column named Location in the format:
City, State

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File Home Transform Add Column View Tools Help

Column From Examples Custom Column Invoke Custom Function Duplicate Column General

Conditional Column Index Column Format Merge Columns Extract Parse From Text From Number From Date & Time

Statistics Standard Scientific Trigonometry Rounding Information Date Time Duration

Queries [3] List of Orders Order Details Sales target

Table.ReorderColumns("#Customer Name Capitalized Each Word",{"Order ID", "Order Date", "CustomerName", "City", "State", "Location"})

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

first row as headers

Changed Type

Kept First 500 Rows

OrderDate changed to DATE ...

Customer Name Capitalized E...

Reordered Columns

Order ID	Order Date	CustomerName	City	State
1 B-25601	01-04-2018	Rohar	Ahmedabad	Gujarat
2 B-25602				
3 B-25603				
4 B-25604				
5 B-25605				
6 B-25606				
7 B-25607				
8 B-25608				
9 B-25609				
10 B-25610				
11 B-25611				
12 B-25612				
13 B-25613				
14 B-25614				
15 B-25615				
16 B-25616				
17 B-25617	17-04-2018	Sagar	Kohima	Nagaland
18 B-25618	18-04-2018	Manju	Hyderabad	Andhra Pradesh
19 B-25619	18-04-2018	Ramesh	Ahmedabad	Gujarat
20 B-25620	20-04-2018	Sarita	Pune	Maharashtra
21 B-25621	20-04-2018	Deepak	Bhopal	Madhya Pradesh
22 B-25622	22-04-2018	Monisha	Jaipur	Rajasthan
23 B-25623	22-04-2018	Atharv	Kolkata	West Bengal
24 B-25624	22-04-2018	Vini	Bangalore	Karnataka

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Column From Examples Custom Column Invoke Custom Function Duplicate Column Conditional Column Index Column Merge Columns Format Extract Parse Statistics Standard Scientific Trigonometry Rounding Date Time Duration

Queries [3] List of Orders Order Details Sales target

Formula Bar: = Table.AddColumn("#Reordered Columns", "Location", each Text.Combine({[City], [State]}, " "), type text)

	Order Date	CustomerName	City	State	Location
1	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat
2	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra
3	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh
4	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur, Rajasthan
5	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata, West Bengal
6	06-04-2018	Hazel	Bangalore	Karnataka	Bangalore, Karnataka
7	06-04-2018	Sonakshi	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir
8	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu
9	09-04-2018	Jitesh	Lucknow	Uttar Pradesh	Lucknow, Uttar Pradesh
10	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar
11	11-04-2018	Anita	Thiruvananthapuram	Kerala	Thiruvananthapuram, Kerala
12	12-04-2018	Shrichand	Chandigarh	Punjab	Chandigarh, Punjab
13	12-04-2018	Mukesh	Chandigarh	Haryana	Chandigarh, Haryana
14	13-04-2018	Vandana	Simla	Himachal Pradesh	Simla, Himachal Pradesh
15	15-04-2018	Bhavna	Gangtok	Sikkim	Gangtok, Sikkim
16	15-04-2018	Kanak	Goa	Goa	Goa, Goa
17	17-04-2018	Sagar	Kohima	Nagaland	Kohima, Nagaland
18	18-04-2018	Manju	Hyderabad	Andhra Pradesh	Hyderabad, Andhra Pradesh
19	18-04-2018	Ramesh	Ahmedabad	Gujarat	Ahmedabad, Gujarat
20	20-04-2018	Sarita	Pune	Maharashtra	Pune, Maharashtra
21	20-04-2018	Deepak	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh
22	22-04-2018	Monisha	Jaipur	Rajasthan	Jaipur, Rajasthan
23	22-04-2018	Atharv	Kolkata	West Bengal	Kolkata, West Bengal
24					

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 03:46 PM

Query Settings

PROPERTIES

Name: List of Orders

APPLIED STEPS

- Source
- first row as headers
- Changed Type
- Kept First 500 Rows
- OrderDate changed to DATE ...
- Customer Name Capitalized E...
- Reordered Columns
- Merged City& State Column

4.2. Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

Custom Column

Add a column that is computed from the other columns.

New column name
Profit Margin

Custom column formula ⓘ
= [Profit]/[Amount]

Available columns
Order ID
Amount
Profit
Quantity
Category
Sub-Category

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Percentage Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [3] List of Orders Order Details Sales target

Order Details

Table: ReorderColumns(*Changed % Type in Profit Margin,{"Order ID", "Amount", "Profit", "Profit Margin", "Quantity", "Category", "Sub-Category"})

Order ID	Amount	Profit	% Profit Margin	Quantity	Category	Sub-Category
1 B-25601	1,275.00	-1,148.00	-90.04%	7	Furniture	Bookcases
2 B-25601	66.00	-12.00	-18.18%	5	Clothing	Stole
3 B-25601	8.00	-2.00	-25.00%	3	Clothing	Hankerchief
4 B-25601	80.00	-56.00	-70.00%	4	Electronics	Electronic Games
5 B-25602	168.00	-111.00	-66.07%	2	Electronics	Phones
6 B-25602	424.00	-272.00	-64.15%	5	Electronics	Phones
7 B-25602	2,617.00	1,151.00	43.98%	4	Electronics	Phones
8 B-25602	561.00	212.00	37.79%	3	Clothing	Saree
9 B-25602	119.00	-5.00	-4.20%	8	Clothing	Saree
10 B-25603	1,355.00	-60.00	-4.43%	5	Clothing	Trousers
11 B-25603	24.00	-30.00	-125.00%	1	Furniture	Chairs
12 B-25603	193.00	-166.00	-86.01%	3	Clothing	Saree
13 B-25603	180.00	5.00	2.78%	3	Clothing	Trousers
14 B-25603	116.00	16.00	13.79%	4	Clothing	Stole
15 B-25603	107.00	36.00	33.64%	6	Clothing	Stole
16 B-25603	12.00	1.00	8.33%	2	Clothing	Hankerchief
17 B-25603	38.00	18.00	47.37%	1	Clothing	Kurti
18 B-25604	65.00	17.00	26.15%	2	Clothing	T-shirt
19 B-25604	157.00	5.00	3.18%	9	Clothing	Saree
20 B-25605	75.00	0.00	0.00%	7	Clothing	Saree
21 B-25606	87.00	4.00	4.60%	2	Clothing	Shirt
22 B-25607	50.00	15.00	30.00%	4	Clothing	Leggings
23 B-25608	1,364.00	-1,864.00	-136.66%	5	Furniture	Tables
24 B-25608	476.00	0.00	0.00%	3	Furniture	Chairs
25 B-25608	257.00	23.00	8.95%	5	Clothing	Hankerchief
26 B-25608	856.00	385.00	44.98%	6	Electronics	Printers
27 B-25609	485.00	29.00	5.98%	4	Electronics	Electronic Games
28 B-25609	25.00	-5.00	-20.00%	4	Clothing	Saree
29 B-25610	1,076.00	-38.00	-3.53%	4	Electronics	Printers
30 B-25610	107.00	-54.00	-50.47%	4	Clothing	Stole
31 B-25610	68.00	-55.00	-80.88%	5	Electronics	Accessories
32 B-25610	781.00	-594.00	-76.06%	6	Electronics	Printers
33 B-25610	43.00	0.00	0.00%	3	Clothing	Saree
34 B-25610	30.00	-5.00	-16.67%	2	Furniture	Furnishings
35 B-25611	160.00	-59.00	-36.88%	2	Clothing	Saree
36 B-25612	259.00	-55.00	-21.24%	2	Furniture	Chairs
37 B-25613	1,603.00	0.00	0.00%	9	Clothing	Saree
38 B-25614	494.00	54.00	10.93%	4	Furniture	Bookcases
39 B-25614	98.00	-12.00	-12.34%	2	Electronics	Electronic Games

7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted headers

Changed column type

Amount Changed Type to Fix...

Profit Margin column Added

Changed % Type in Profit Mar...

Reordered Profit Margin Colu...

PREVIEW DOWNLOADED AT 04:26 PM

35°C Mostly sunny 04:27 PM 13-07-2026

The **Customer Name** column was formatted to **Proper Case** to standardize text values and improve data consistency. A new **Location** column was created by merging the **City** and **State** columns into a single field in the format **City, State**, making geographical analysis easier.

A custom column named **Profit Margin** was created using the formula:

Profit Margin = Profit ÷ Amount

The result was formatted as a percentage to measure the profitability of each order.

5. Conditional Column

5.1. Add a conditional column named Profit Status based on:

5.1.1. Profit < 0 → Loss , Profit = 0 → Break-Even, Profit > 0 → Profit

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Profit Status

	Column Name	Operator	Value ①		Output ①
If	Profit	is less than	0	Then	Loss
Else If	Profit	equals	0	Then	Break-Even

Add Clause

Else ①
Profit

OK Cancel

A conditional column named **Profit Status** was added using the following business rules:

- **Profit < 0 → Loss**
- **Profit = 0 → Break-Even**
- **Profit > 0 → Profit**

This categorizes each order based on its financial performance.

6. Merging Data (Joins)

6.1. Merge the List of Orders and Order Details tables using Order ID.

Merge

Select tables and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	City	State	Location
B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat
B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra
B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh
B-25604	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur, Rajasthan
B-25605	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata, West Bengal

Order Details

Order ID	Amount	Profit	Profit Margin	Quantity	Category	Sub-Category	Profit Status
B-25601	1,275.00	-1,148.00	-90.04%	7	Furniture	Bookcases	Loss
B-25601	66.00	-12.00	-18.18%	5	Clothing	Stole	Loss
B-25601	8.00	-2.00	-25.00%	3	Clothing	Hankerchief	Loss
B-25601	80.00	-56.00	-70.00%	4	Electronics	Electronic Games	Loss
B-25602	168.00	-111.00	-66.07%	2	Electronics	Phones	Loss

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table.

OK

Cancel

6.2. Name the merged output table as Orders Data.

The screenshot displays the Microsoft Power BI Desktop interface. The main window shows a data table with 14 columns and 999 rows. The columns are: Order ID, Order Date, Customer Name, City, State, Location, Order Details Order ID, Order Details Amount, and Order Details Profit. The data is a result of a merge operation between 'List of Orders' and 'Order Details' tables, joined on the 'Order ID' column using a Left Outer Join. The merged table is named 'Orders Data' in the 'Queries' pane on the left. The 'Query Settings' pane on the right shows the 'Expanded Order Details' source. The status bar at the bottom indicates '14 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Order ID	Order Date	Customer Name	City	State	Location	Order Details Order ID	Order Details Amount	Order Details Profit
1	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat	B-25601	1,275.00	
2	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat	B-25601	66.00	
3	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat	B-25601	8.00	
4	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad, Gujarat	B-25601	80.00	
5	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra	B-25602	168.00	
6	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra	B-25602	424.00	
7	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra	B-25602	2,617.00	
8	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra	B-25602	561.00	
9	01-04-2018	Pearl	Pune	Maharashtra	Pune, Maharashtra	B-25602	119.00	
10	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	1,355.00	
11	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	24.00	
12	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	193.00	
13	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	180.00	
14	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	116.00	
15	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	107.00	
16	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	12.00	
17	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-25603	38.00	
18	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur, Rajasthan	B-25604	65.00	
19	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur, Rajasthan	B-25604	157.00	
20	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata, West Bengal	B-25605	75.00	
21	06-04-2018	Hazel	Bangalore	Karnataka	Bangalore, Karnataka	B-25606	87.00	
22	06-04-2018	Sonakshi	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-25607	50.00	
23	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	1,364.00	
24	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	476.00	
25	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	257.00	
26	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	856.00	
27	09-04-2018	Iltesh	Lucknow	Uttar Pradesh	Lucknow, Uttar Pradesh	B-25609	485.00	
28	09-04-2018	Iltesh	Lucknow	Uttar Pradesh	Lucknow, Uttar Pradesh	B-25609	25.00	
29	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	1,076.00	
30	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	107.00	
31	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	68.00	
32	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	781.00	
33	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	43.00	
34	09-04-2018	Yogesh	Patna	Bihar	Patna, Bihar	B-25610	30.00	
35	11-04-2018	Anita	Thiruvananthapuram	Kerala	Thiruvananthapuram, Kerala	B-25611	160.00	
36	12-04-2018	Shrichand	Chandigarh	Punjab	Chandigarh, Punjab	B-25612	259.00	
37	12-04-2018	Mukesh	Chandigarh	Haryana	Chandigarh, Haryana	B-25613	1,603.00	
38	13-04-2018	Vandana	Simla	Himachal Pradesh	Simla, Himachal Pradesh	B-25614	494.00	

The **List of Orders** and **Order Details** tables were merged using the **Order ID** column with a **Left Outer Join**. The merged table was named **Orders Data**, providing a consolidated dataset containing both customer information and order details.

7. Handling Missing Data & Duplicate Data

- **List of Order** dataset contains 60 blank rows.
- **Already Restrict the List of Orders table to the first 500 rows.**
- **Now no null values in the dataset**
- Order Details & Sales Target datasets have No missing values and No action required.
- No Duplicate in all datasets

8. Sorting and Filtering Data

8.1. Sort records in the Orders Data table by Order Date in descending order.

The screenshot displays the Power BI Desktop interface. The main view shows a table with 14 columns and 999+ rows. The columns are: Order ID, Order Date, CustomerName, City, State, Location, Order Details.Order ID, Order Details.Amount, and Order Details. The table is sorted by Order Date in descending order. The first row shows an order from 31-05-2019, and the last row shows an order from 27-03-2019. The right-hand pane shows the 'Query Settings' for the 'Orders Data' query, with the 'APPLIED STEPS' list containing 'Sort Order Date in descending order...'. The bottom status bar indicates '14 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Order ID	Order Date	CustomerName	City	State	Location	Order Details.Order ID	Order Details.Amount
B-26100	31-05-2019	Hitika	Indore	Madhya Pradesh	Indore, Madhya Pradesh	B-26100	72.00
B-26100	31-05-2019	Hitika	Indore	Madhya Pradesh	Indore, Madhya Pradesh	B-26100	828.00
B-26100	31-05-2019	Hitika	Indore	Madhya Pradesh	Indore, Madhya Pradesh	B-26100	34.00
B-26099	30-03-2019	Bhishm	Mumbai	Maharashtra	Mumbai, Maharashtra	B-26099	207.00
B-26099	30-03-2019	Bhishm	Mumbai	Maharashtra	Mumbai, Maharashtra	B-26099	2,366.00
B-26099	30-03-2019	Bhishm	Mumbai	Maharashtra	Mumbai, Maharashtra	B-26099	9.00
B-26099	30-03-2019	Bhishm	Mumbai	Maharashtra	Mumbai, Maharashtra	B-26099	835.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	46.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	497.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	409.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	59.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	96.00
B-26098	29-03-2019	Pinky	Kashmir	Jammu and Kashmir	Kashmir, Jammu and Kashmir	B-26098	82.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	103.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	97.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	19.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	88.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	39.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	264.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	451.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	185.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	14.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	671.00
B-26095	28-03-2019	Monisha	Jaipur	Rajasthan	Jaipur, Rajasthan	B-26095	6.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	45.00
B-26096	28-03-2019	Atharv	Kolkata	West Bengal	Kolkata, West Bengal	B-26096	140.00
B-26097	28-03-2019	Vini	Bangalore	Karnataka	Bangalore, Karnataka	B-26097	663.00
B-26090	27-03-2019	Sagar	Kohima	Nagaland	Kohima, Nagaland	B-26090	80.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	148.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	492.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	33.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	49.00
B-26091	27-03-2019	Manju	Hyderabad	Andhra Pradesh	Hyderabad, Andhra Pradesh	B-26091	158.00
B-26094	27-03-2019	Deepak	Bhopal	Madhya Pradesh	Bhopal, Madhya Pradesh	B-26094	152.00
B-26092	27-03-2019	Ramesh	Ahmedabad	Gujarat	Ahmedabad, Gujarat	B-26092	97.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	852.00
B-26091	27-03-2019	Manju	Hyderabad	Andhra Pradesh	Hyderabad, Andhra Pradesh	B-26091	59.00
B-26093	27-03-2019	Sarita	Pune	Maharashtra	Pune, Maharashtra	B-26093	2,847.00

8.2. Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Transpose Reverse Rows Detect Data Type Rename

Replace Values Fill

Unpivot Columns Move Convert to List

Any Column

Split Column Format Text Column

Merge Columns Extract Parse

Statistics Standard Scientific Information

Number Column

Trigonometry Rounding

Date Time Duration

Date & Time Column

Run R script Run Python script

Scripts

Queries [4]

List of Orders Order Details Sales target Orders Data

fx = Table.SelectRows("#Sort Order Date in descending order", each ([State] = "Tamil Nadu"))

	Order ID	Order Date	CustomerName	City	State	Location	Order Details.Order ID	Order Details.Amount	Order Details
1	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	79.00	
2	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	169.00	
3	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	359.00	
4	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	93.00	
5	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	24.00	
6	B-26081	22-03-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26081	637.00	
7	B-26018	14-02-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26018	61.00	
8	B-26018	14-02-2019	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26018	326.00	
9	B-26008	09-02-2019	Kalyani	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26008	22.00	
10	B-26008	09-02-2019	Kalyani	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26008	206.00	
11	B-26008	09-02-2019	Kalyani	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26008	10.00	
12	B-26008	09-02-2019	Kalyani	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-26008	57.00	
13	B-25860	15-11-2018	Akshay	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25860	112.00	
14	B-25788	21-09-2018	Dinesh	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25788	12.00	
15	B-25716	11-07-2018	Surabhi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25716	58.00	
16	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	207.00	
17	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	27.00	
18	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	65.00	
19	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	87.00	
20	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	7.00	
21	B-25698	23-06-2018	Amisha	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25698	516.00	
22	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	257.00	
23	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	856.00	
24	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	476.00	
25	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai, Tamil Nadu	B-25608	1,364.00	

Query Settings

PROPERTIES

Name

Orders Data

All Properties

APPLIED STEPS

Source

Expanded Order Details

Sort Order Date in descending or...

Filtered Rows by State TN

9. Grouping and Aggregating Data

9.1. Duplicate the Order Details table

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply Close New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Text Use First Row as Headers Replace Values Transform Merge Queries Append Queries Combine Files Combine

Queries [5]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Duplicate Order Details**

fx = Table.AddColumn(#'Reordered Profit Margin Columns', 'Profit Status', each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else "Profit")

Order ID	Amount	Profit	Profit Margin	Quantity	Category	Sub-Category	Profit Status
1462 B-26091	258.00	89.00	43.87%	3	Clothing	Stole	Profit
1463 B-26091	29.00	10.00	34.48%	4	Clothing	Hankerchief	Profit
1464 B-26091	59.00	10.00	16.95%	4	Clothing	Leggings	Profit
1465 B-26092	97.00	14.00	14.43%	2	Clothing	T-shirt	Profit
1466 B-26093	33.00	-1.00	-3.03%	1	Clothing	Saree	Loss
1467 B-26093	2,847.00	712.00	25.01%	8	Electronics	Printers	Profit
1468 B-26093	852.00	51.00	5.99%	5	Furniture	Bookcases	Profit
1469 B-26093	492.00	187.00	38.01%	2	Electronics	Accessories	Profit
1470 B-26093	81.00	41.00	50.62%	5	Clothing	Leggings	Profit
1471 B-26093	49.00	5.00	10.20%	4	Clothing	Hankerchief	Profit
1472 B-26093	148.00	25.00	16.89%	3	Clothing	Saree	Profit
1473 B-26094	152.00	50.00	32.89%	6	Clothing	Stole	Profit
1474 B-26095	6.00	1.00	16.67%	1	Clothing	Kurti	Profit
1475 B-26096	45.00	9.00	20.00%	3	Clothing	Leggings	Profit
1476 B-26096	103.00	46.00	44.66%	2	Clothing	Saree	Profit
1477 B-26096	140.00	56.00	40.00%	4	Clothing	Shirt	Profit
1478 B-26096	88.00	11.00	12.50%	3	Electronics	Accessories	Profit
1479 B-26096	451.00	25.00	5.54%	3	Electronics	Phones	Profit
1480 B-26096	264.00	-26.00	-9.85%	3	Clothing	Trousers	Loss
1481 B-26097	97.00	12.00	12.37%	2	Clothing	Hankerchief	Profit
1482 B-26097	14.00	5.00	35.71%	1	Clothing	Hankerchief	Profit
1483 B-26097	19.00	8.00	42.11%	2	Clothing	Hankerchief	Profit
1484 B-26097	39.00	18.00	46.15%	2	Clothing	Leggings	Profit
1485 B-26097	185.00	-26.00	-14.05%	6	Furniture	Chairs	Loss
1486 B-26097	663.00	-212.00	-31.98%	5	Electronics	Printers	Loss
1487 B-26097	671.00	-309.00	-46.05%	5	Electronics	Electronic Games	Loss
1488 B-26098	82.00	8.00	9.76%	3	Electronics	Accessories	Profit
1489 B-26098	497.00	179.00	36.02%	3	Furniture	Chairs	Profit
1490 B-26098	96.00	48.00	50.00%	5	Clothing	Leggings	Profit
1491 B-26098	409.00	86.00	21.03%	3	Clothing	Saree	Profit
1492 B-26098	59.00	15.00	25.42%	2	Clothing	T-shirt	Profit
1493 B-26098	46.00	14.00	30.43%	5	Clothing	Skirt	Profit
1494 B-26099	9.00	3.00	33.33%	1	Clothing	Skirt	Profit
1495 B-26099	207.00	37.00	17.87%	4	Clothing	Hankerchief	Profit
1496 B-26099	835.00	267.00	31.98%	5	Electronics	Phones	Profit
1497 B-26099	2,366.00	552.00	23.33%	5	Clothing	Trousers	Profit
1498 B-26100	828.00	230.00	27.78%	2	Furniture	Chairs	Profit
1499 B-26100	34.00	10.00	29.41%	2	Clothing	T-shirt	Profit
1500 B-26100	72.00	16.00	22.22%	2	Clothing	Shirt	Profit

8 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name Duplicate Order Details

APPLIED STEPS

- Source
- Promoted headers
- Changed column type
- Amount Changed Type to Fix...
- Profit Margin column Added
- Changed % Type in Profit Mar...
- Reordered Profit Margin Colu...
- Added Profit Status Condition...**

30°C Partly cloudy 08:10 PM 13-07-2026

9.1.1. Count of Order ID

×

Group By

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Order ID

New column name

Count

Operation

Count Rows

Column

OK

Cancel

✕ ✓ <i>f</i>_x = Table.Group(#"Added Profit Status Conditional Column", {"Order ID"}, {"Count", each Table.RowCount(_), Int64.Type})		
	Order ID	Count
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2
19	B-25619	1
20	B-25620	1
21	B-25621	3
22	B-25622	1

9.1.2. Average Profit by Category

Power BI Assignment 1 – Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Remove Rows Sort Split Column Group By Data Type: Decimal Number Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Duplicate Order Details
- Duplicate Sales target

fx = Table.Group(#"Added Profit Status Conditional Column", {"Category"}, {{ "Average Profit", each List.Average([Profit]), type nullable number}})

	Category	Average Profit
1	Furniture	9.456790123
2	Clothing	11.76290832
3	Electronics	34.07142857

9.1.3. Total Amount by Sub-Category

Power BI Desktop - Data Transformation & Data Modeling

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Choose Columns

Queries [6]

- List of Orders
- Order Details
- Sales target
- Orders Data
- Duplicate Order Details**
- Duplicate Sales target

fx = Table.Group(#"Added Profit Status Conditional Column", {"Sub-Category", "Total Amount"}, {

	Sub-Category	Total Amount
1	Bookcases	56861
2	Stole	18546
3	Hankerchief	14608
4	Electronic Games	39168
5	Phones	46119
6	Saree	53511
7	Trousers	30039
8	Chairs	34222
9	Kurti	3361
10	T-shirt	7382
11	Shirt	7555
12	Leggings	2106
13	Tables	22614
14	Printers	58252
15	Accessories	21728
16	Furnishings	13484
17	Skirt	1946

9.2. Duplicate the Sales Target table and aggregate the total Target by Month of order date.

The screenshot shows the Power BI Desktop interface with the 'Transform' tab selected. The 'Queries' pane on the left lists several queries, with 'Duplicate Sales target' highlighted. The main view displays a table with the following data:

	Month of Order Date	1.2 Month of order date.1
1	18-04-2026	31400
2	18-05-2026	31500
3	18-06-2026	31600
4	18-07-2026	33800
5	18-08-2026	33900
6	18-09-2026	34000
7	18-10-2026	36100
8	18-11-2026	36300
9	18-12-2026	36400
10	19-01-2026	43500
11	19-02-2026	43600
12	19-03-2026	43800

10.Data Modeling

10.1. Establish a relationship between:

10.1.1. List of Orders and Order Details using Order ID

Edit relationship

×

Select tables and columns that are related.

From table

Order Details

Category	Order ID	Profit Margin	Profit Status	Quantity	Sub-Category
Clothing	B-25603	0.08333333333...	Profit	2	Hankerchief
Clothing	B-25608	0.0894941634...	Profit	5	Hankerchief
Clothing	B-25615	0.2941176470...	Profit	5	Hankerchief

To table

List of Orders

City	CustomerName	Location	Order Date	Order ID	State
Indore	Hitika	Indore, Madh...	23 April 2018	B-25627	Madhya Prad...
Indore	Ashmi	Indore, Madh...	26 April 2018	B-25637	Madhya Prad...
Indore	Yaanvi	Indore, Madh...	01 May 2018	B-25645	Madhya Prad...

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

10.2. Establish a relationship between:

10.2.1. Order Details and Sales Target using Category

New relationship

×

Select tables and columns that are related.

From table

Order Details

Category	Order ID	Profit Margin	Profit Status	Quantity	Sub-Category
Clothing	B-25603	0.08333333333...	Profit	2	Hankerchief
Clothing	B-25608	0.0894941634...	Profit	5	Hankerchief
Clothing	B-25615	0.2941176470...	Profit	5	Hankerchief

To table

Sales target

Category	Month of Ord...
Furniture	18 April 2026
Furniture	18 May 2026
Furniture	18 June 2026

Cardinality

Many to many (*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

10.3. Use Manage Relationships to ensure relationships are active and correctly defined.

